## *Problems*

**2.1.** Use the method of separation of variables to solve the following initial-Neumann problem:

$$\begin{cases} u_t - u_{xx} = 0 & 0 < x < L, t > 0 \\ u(x,0) = x & 0 < x < L \\ u_x(0,t) = u_x(L,t) = 0 & t > 0. \end{cases}$$

**2.2.** Use the method of separation of variables to solve the following non homogeneous initial-Neumann problem:

$$\begin{cases} u_t - u_{xx} = tx & 0 < x < \pi, t > 0 \\ u(x,0) = 1 & 0 \le x \le \pi \\ u_x(0,t) = u_x(L,t) = 0 & t > 0. \end{cases}$$

[*Hint*: Write the candidate solution as $u(x,t) = \sum_{k \ge 0} c_k(t) v_k(x)$ where $v_k$ are the eigenfunctions of the eigenvalue problem associated with the homogeneous equation].

**2.3.** Use the method of separation of variables to solve (at least formally) the following mixed problem:

$$\begin{cases} u_t - Du_{xx} = 0 & 0 < x < \pi, t > 0 \\ u(x,0) = g(x) & 0 \le x \le \pi \\ u_x(0,t) = 0 & \\ u_x(L,t) + u(L,t) = U & t > 0. \end{cases}$$

[*Answer*: $u(x,t) = \sum_{k \ge 0} c_k e^{-D\mu_k^2 t} \cos \mu_k x$, where the numbers $\mu_k$ are the positive solutions of the equation $\mu \tan \mu = 1$].

**2.4.** Prove that, if $w_t - D\Delta w = 0$ in $Q_T$ and $w \in C\left(\overline{Q}_T\right)$, then

$$\min_{\partial_p Q_T} w \le w(\mathbf{x},t) \le \max_{\partial_p Q_T} w \qquad \textit{for every } (\mathbf{x},t) \in Q_T.$$

**2.5.** Prove Corollary 2.1.

[*Hint*: b). Let $u = v - w$, $M = \max_{\overline{Q}_T} |f_1 - f_2|$ and apply Theorem 2.2 to $z_\pm = \pm u - Mt$].

**2.6.** Let $g(t) = M$ for $0 \le t \le 1$ and $g(t) = M - (1-t)^4$ for $1 < t \le 2$. Let $u$ be the solution of $u_t - u_{xx} = 0$ in $Q_2 = (0,2) \times (0,2)$, $u = g$ on $\partial_p Q_2$. Compute $u(1,1)$ and check that it is the maximum of $u$. Is this in contrast with the strong maximum principle of Remark 2.4?

**2.7.** Suppose $u = u(x,t)$ is a solution of the heat equation in a plane domain $D_T = Q_T \backslash \left(\bar{Q}_1 \cup \bar{Q}_2\right)$ where $Q_1$ and $Q_2$ are the rectangles in figure 2.19. Assume that $u$ attains its maximum $M$ at the interior point $(x_1, t_1)$. Where else $u = M$?

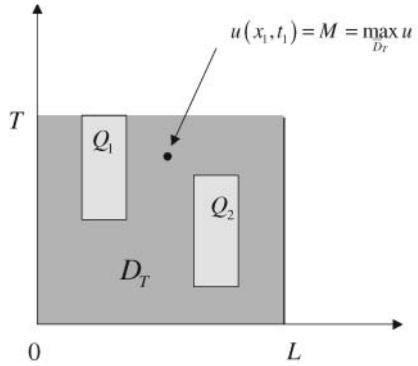


**Fig. 2.19.** At which points $(x,t)$, $u(x,t) = M$?

**2.8.** Find the similarity solutions of the equation $u_t - u_{xx} = 0$ of the form $u(x,t) = U\left(x/\sqrt{t}\right)$ and express the result in term of the *error function*

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-z^2} dz.$$

Find the solution of the problem $u_t - u_{xx} = 0$ in $x > 0, t > 0$ satisfying the conditions $u(0,t) = 1$ and $u(x,0) = 0$, $x > 0$.

**2.9.** Determine for which $\alpha$ and $\beta$ there exist similarity solutions to $u_t - u_{xx} = f(x)$ of the form $t^\alpha U\left(x/t^\beta\right)$ in each one of the following cases:

$(a)\ f(x) = 0, \quad (b)\ f(x) = 1, \quad (c)\ f(x) = x.$

[*Answer*: $(a)$ $\alpha$ arbitrary, $\beta = 1/2$. $(b)$ $\alpha = 1$, $\beta = 1/2$. $(c)$ $\alpha = 3/2$, $\beta = 1/2$].

**2.10.** (*Reflecting barriers and Neumann condition*). Consider the symmetric random walk of Section 2.4. Suppose that a perfectly *reflecting* barrier is located at the point $L = \overline{m}h + \frac{h}{2} > 0$. By this we mean that if the particle hits the point $L - \frac{h}{2}$ at time $t$ and moves to the right, then it is reflected and it comes back to $L - \frac{h}{2}$ at time $t + \tau$. Show that when $h, \tau \to 0$ and $h^2/\tau = 2D$, $p = p(x,t)$ is a solution of the problem

$$\begin{cases} p_t - Dp_{xx} = 0 & x < L, t > 0 \\ p(x,0) = \delta & x < L \\ p_x(L,t) = 0 & t > 0 \end{cases}$$

and moreover $\int_{-\infty}^{L} p(x,t)\, dx = 1$. Compute explicitly the solution.
[*Answer*: $p(x,t) = \Gamma_D(x,t) + \Gamma_D(x - 2L, t)$].

**2.11.** (*Absorbing barrires and Dirichlet condition*). Consider the symmetric random walk of Section 2.4. Suppose that a perfectly *absorbing* barrier is located at the point $L = \overline{m}h > 0$. By this we mean that if the particle hits the point $L - h$

at time $t$ and moves to the right then it is absorbed and stops at $L$. Show that when $h, \tau \to 0$ and $h^2/\tau = 2D$, $p = p(x,t)$ is a solution of the problem

$$\begin{cases} p_t - Dp_{xx} = 0 & x < L, t > 0 \\ p(x,0) = \delta & x < L \\ p(L,t) = 0 & t > 0 \end{cases}$$

Compute explicitly the solution.

[*Answer*: $p(x,t) = \Gamma_D(x,t) - \Gamma_D(x-2L,t)$.]

**2.12.** Use the partial Fourier transform $\hat{u}(\xi,t) = \int_{\mathbb{R}} e^{-ix\xi} u(x,t)\,dx$ to solve the global Cauchy problem (2.129) and rediscover formula (2.130).

**2.13.** Prove Theorem 2.5 under the condition

$$z(x,t) \le C, \ \ x \in \mathbb{R}, 0 \le t \le T,$$

using the following steps.

a) Let $\sup_{\mathbb{R}} \ z(x,0) = M_0$ and define

$$w(x,t) = \frac{2C}{L^2}(\frac{x^2}{2} + Dt) + M_0.$$

Check that $w_t - Dw_{xx} = 0$ and use the maximum principle to show that $w \ge z$ in the rectangle $R_L = [-L, L] \times [0,T]$.

b) Fix an arbitrary point $(x_0, t_0)$ and choose $L$ large enough to have $(x_0, t_0) \in R_L$. Using a) deduce that $z(x_0, t_0) \le M_0$.

**2.14.** Find an explicit formula for the solution of the global Cauchy problem

$$\begin{cases} u_t = Du_{xx} + bu_x + cu & x \in \mathbb{R}, t > 0 \\ u(x,0) = g(x) & x \in \mathbb{R}. \end{cases}$$

where $D, b, c$ are constant coefficients. Show that, if $c < 0$ and $g$ is bounded, $u(x,t) \to 0$ as $t \to +\infty$

[*Hint*: Choose $h, k$ such that $v(x,t) = u(x,t)\,e^{hx+kt}$ is a solution of $v_t = Dv_{xx}$].

**2.15.** Find an explicit formula for the solution of the Cauchy problem

$$\begin{cases} u_t = u_{xx} & x > 0, t > 0 \\ u(x,0) = g(x) & x \ge 0 \\ u(0,t) = 0 & t > 0. \end{cases}$$

with $g$ continuous and $g(0) = 0$.

[*Hint*: Extend $g$ to $x < 0$ by odd reflection: $g(-x) = -g(x)$. Solve the corresponding global Cauchy problem and write the result as an integral on $(0, +\infty)$].

**2.16.** Let $Q_T = \Omega \times (0,T)$, with $\Omega$ bounded domain in $\mathbb{R}^n$. Let $u \in C^{2,1}(Q_T) \cap C(\overline{Q_T})$ satisfy the equation

$$u_t = D\Delta u + \mathbf{b}(\mathbf{x},t) \cdot \nabla u + c(\mathbf{x},t)\,u \qquad \text{in } Q_T$$

where $\mathbf{b}$ and $c$ are continuous in $\overline{Q}_T$. Show that if $u \geq 0$ (resp. $u \leq 0$) on $\partial_p Q_T$ then $u \geq 0$ (resp. $u \leq 0$) in $Q_T$.

[*Hint*: Assume first that $c(\mathbf{x},t) \leq a < 0$. Then reduce to this case by setting $u = ve^{kt}$ with a suitable $k > 0$].

**2.17**. Fill in the details in the arguments of Section 6.2, leading to formulas (2.108) and (2.109).

**2.18**. Solve the following initial-Dirichlet problem in $B_1 = \left\{\mathbf{x} \in \mathbb{R}^3 \colon |\mathbf{x}| < 1\right\}$:

$$\begin{cases} u_t = \Delta u & \mathbf{x} \in B_1, t > 0 \\ u(\mathbf{x}, 0) = 0 & \mathbf{x} \in B_1 \\ u(\boldsymbol{\sigma}, t) = 1 & \boldsymbol{\sigma} \in \partial B_1, t > 0. \end{cases}$$

Compute $\lim_{t\to+\infty} u$.

[*Hint*: The solution is radial so that $u = u(r,t)$, $r = |\mathbf{x}|$. Observe that $\Delta u = u_{rr} + \dfrac{2}{r} u_r = \dfrac{1}{r}(ru)_{rr}$. Let $v = ru$, reduce to homogeneous Dirichlet condition and use separation of variables].

**2.19.** Solve the following initial-Dirichlet problem

$$\begin{cases} u_t = \Delta u & \mathbf{x} \in K, t > 0 \\ u(\mathbf{x}, 0) = 0 & \mathbf{x} \in K \\ u(\boldsymbol{\sigma}, t) = 1 & \boldsymbol{\sigma} \in \partial K, \ t > 0. \end{cases}$$

where $K$ is the rectangular box

$$K = \left\{(x, y, z) \in \mathbb{R}^3 \colon 0 < x < a, 0 < y < b, 0 < z < c\right\}.$$

Compute $\lim_{t\to+\infty} u$.

**2.20.** Solve the following initial-Neumann problem in $B_1 = \left\{\mathbf{x} \in \mathbb{R}^3 \colon |\mathbf{x}| < 1\right\}$:

$$\begin{cases} u_t = \Delta u & \mathbf{x} \in B_1, t > 0 \\ u(\mathbf{x}, 0) = |\mathbf{x}| & \mathbf{x} \in B_1 \\ u_{\boldsymbol{\nu}}(\boldsymbol{\sigma}, t) = 1 & \boldsymbol{\sigma} \in \partial B_1, t > 0. \end{cases}$$

**2.21.** Solve the following non homogeneous initial-Dirichlet problem in the unit sphere $B_1$ $(u = u(r,t), r = |\mathbf{x}|)$:

$$\begin{cases} u_t - (u_{rr} + \dfrac{2}{r} u_r) = qe^{-t} & 0 < r < 1, t > 0 \\ u(r, 0) = U & 0 \leq r \leq 1 \\ u(1, t) = 0 & t > 0. \end{cases}$$

[*Answer*: The solution is

$$u(r,t) = \frac{2}{r} \sum_{n=1}^{\infty} \frac{(-1)^n}{\lambda_n} \sin(\lambda_n r) \left\{ \frac{q}{1 - \lambda_n^2} \left( e^{-t} - e^{-\lambda_n^2 t} \right) - U e^{-\lambda_n^2 t} \right\}$$

where $\lambda_n = n\pi$].

**2.22.** Using the maximum principle, compare the values of two call options $C^{(1)}$ and $C^{(2)}$ in the following cases:

($a$) Same exercise price and $T_1 > T_2$. ($b$) Same expiry time and $E_1 > E_2$.

**2.23.** Justify carefully the orbit configuration of figure 2.18 and in particular that the unstable orbit $\gamma$ connects the two equilibrium points of system (2.175), by filling in the details in the following steps:

**1.** Let $\mathbf{F} = V\mathbf{i} + \left(-cV + U^2 - U\right)\mathbf{j}$ and $\mathbf{n}$ be the interior normal to the boundary of the triangle $\Omega$ in figure 2.20. Show that, if $\beta$ is large enough, $\mathbf{F} \cdot \mathbf{n} > 0$ along $\partial\Omega$.

**2.** Deduce that all the orbits of system (2.175) starting at a point in $\Omega$ cannot leave $\Omega$ (i.e. $\Omega$ is a *positively invariant region*) and converge to the origin as $z \to +\infty$.

**3.** Finally, deduce that the unstable separatrix $\gamma$ of the saddle point $(1, 0)$ approaches $(0, 0)$ as $z \to +\infty$.

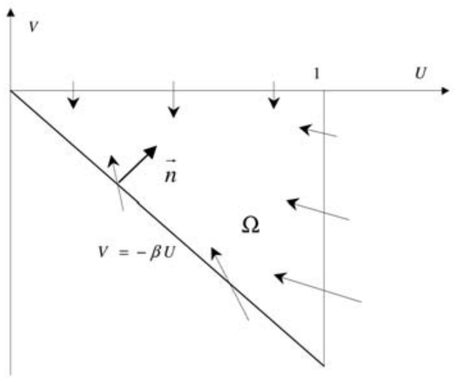


**Fig. 2.20.** Trapping region for the orbits of the vector field $\mathbf{F} = V\mathbf{i} + \left(-cV + U^2 - U\right)\mathbf{j}$